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Course code:CSA0604

Course name: Data Handling and Visualization

Set 16: Customer Demographics Analysis

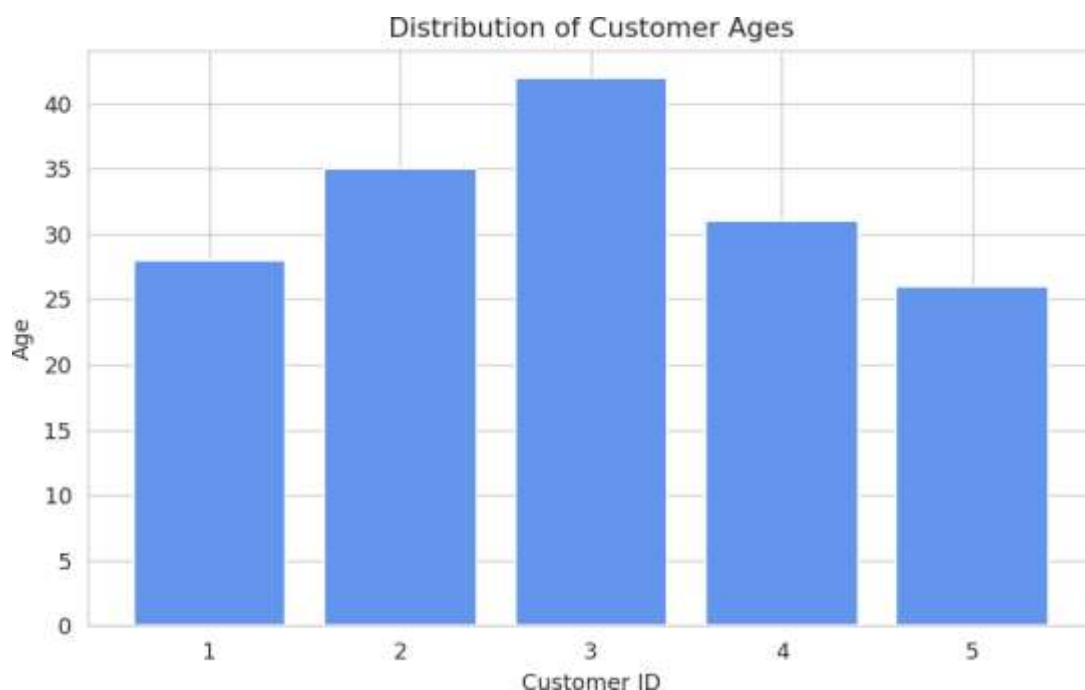
Question 1

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

R Code

```
# 1. Bar chart - distribution of customer ages
ggplot(df, aes(x = factor(CustomerID), y = Age)) +
  geom_bar(stat = "identity", fill = "cornflowerblue") +
  labs(title = "Distribution of Customer Ages", x = "Customer ID", y = "Age") +
  theme_minimal()
ggsave("bar_chart_customer_ages.png", width = 7, height = 5)
```

Output



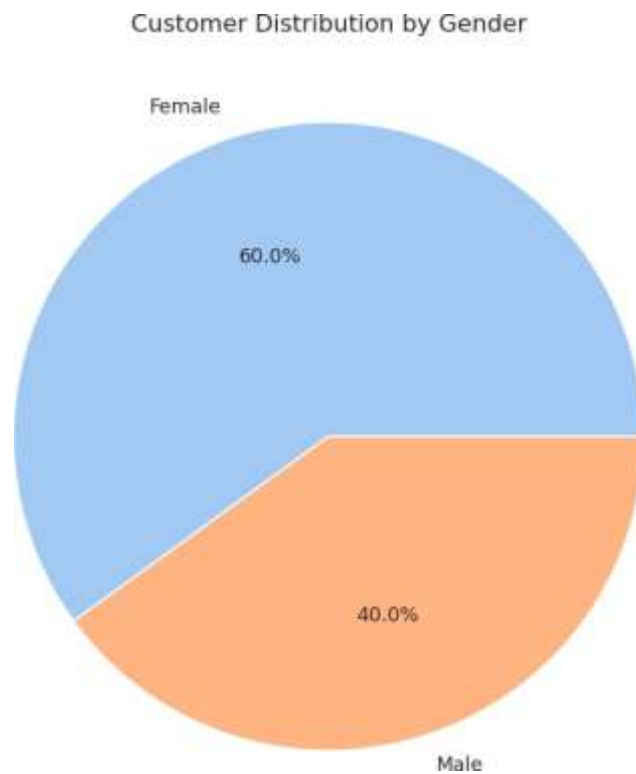
Question 2

Generate a pie chart to represent the distribution of customers by gender.

R Code

```
# 2. Pie chart - distribution of customers by gender
gender_counts <- df %>% count(Gender)
ggplot(gender_counts, aes(x = "", y = n, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Distribution by Gender") +
  theme_void()
ggsave("pie_chart_gender.png", width = 7, height = 5)
```

Output



Question 3

Build a table to show the demographic information for each customer. Label the table elements.

R Code

```
# 3. Table - demographic information for each customer
demo_table <- df %>% gt() %>% tab_header(title = "Customer Demographic Information")
gtsave(demo_table, "table_demographics.png")
print(df)
```

Output

Customer Demographic Information

CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000
4	31	Male	55000
5	26	Female	48000

Set 17: Employee Performance Analysis

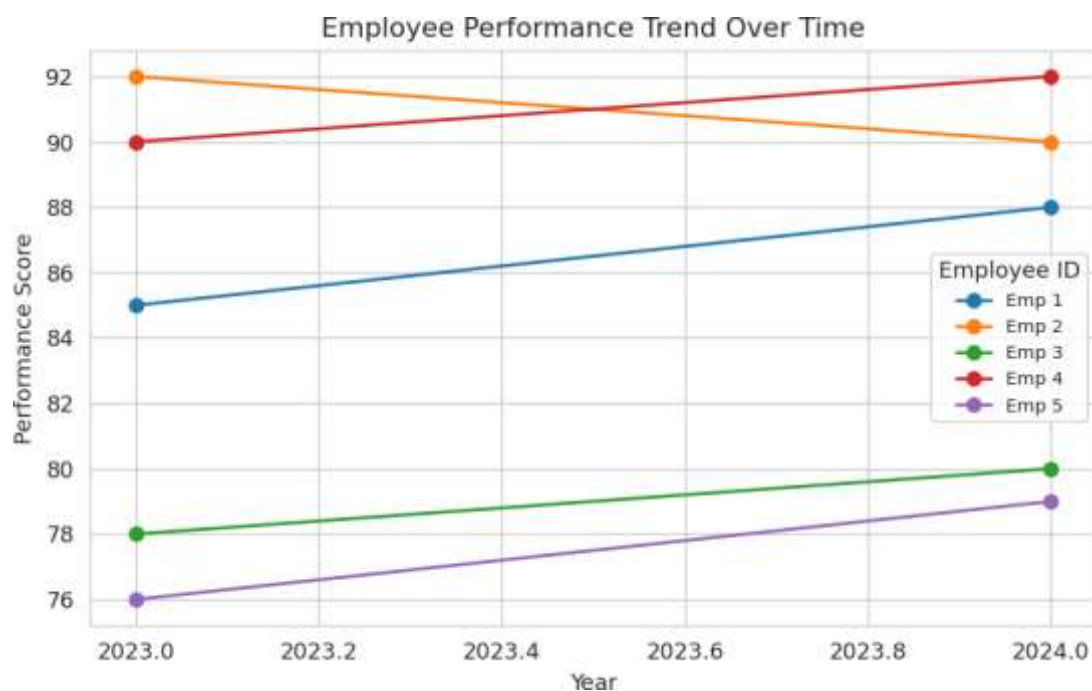
Question 1

Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R Code

```
# 1. Line chart - performance trend of employees over time
ggplot(df, aes(x = Year, y = PerformanceScore, color = factor(EmployeeID), group = EmployeeID)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Employee Performance Trend", x = "Year", y = "Performance Score", color = "Employee ID") +
  theme_minimal()
ggsave("line_chart_performance_trend.png", width = 7, height = 5)
```

Output



Question 2

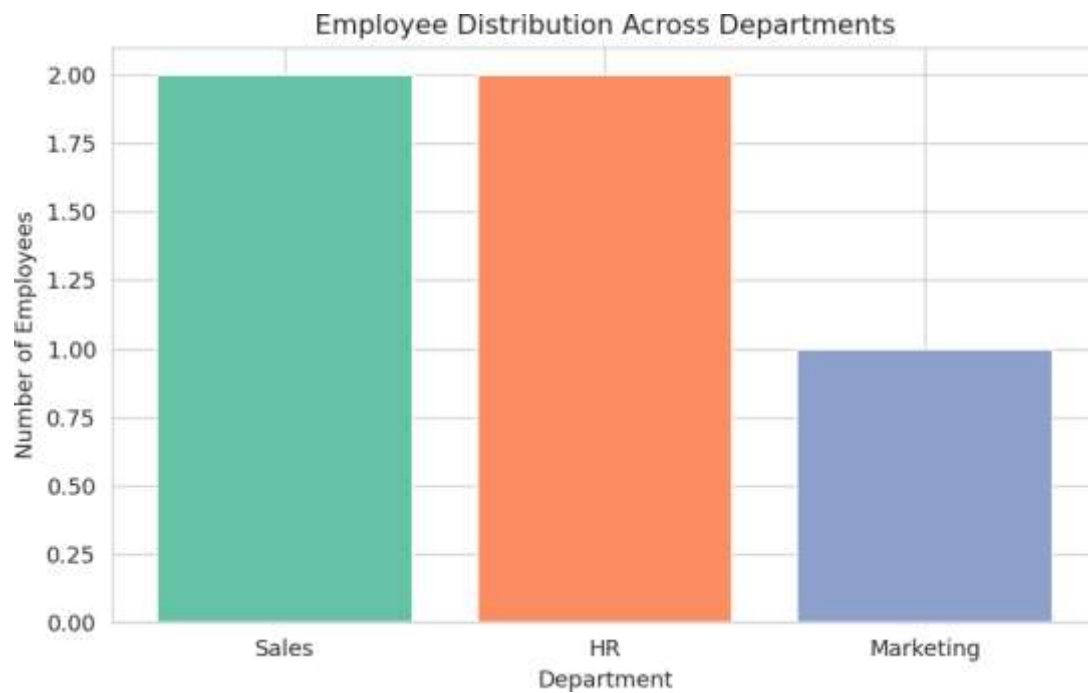
Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Code

```
# 2. Bar chart - distribution of employees across departments
dept_counts <- df %>% distinct(EmployeeID, Department) %>% count(Department)
ggplot(dept_counts, aes(x = Department, y = n, fill = Department)) +
```

```
geom_bar(stat = "identity") +
labs(title = "Employee Distribution by Department", x = "Department", y = "Count") +
theme_minimal()
ggsave("bar_chart_department_distribution.png", width = 7, height = 5)
```

Output



Question 3

Build a table to display the performance data for each employee. Label the table elements.

R Code

```
# 3. Table - performance data for each employee
perf_table <- df %>% gt() %>% tab_header(title = "Employee Performance Data")
gtsave(perf_table, "table_employee_performance.png")
print(df)
```

Output

EmployeeID	Department	YearsOfService	PerformanceScore	Year
1	Sales	5	85	2023
2	HR	3	92	2023
3	Marketing	7	78	2023
4	Sales	4	90	2023
5	HR	2	76	2023

Set 18: Product Inventory Management

Question 1

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code

```
# 1. Bar chart - quantity of each product in inventory
ggplot(df, aes(x = ProductName, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity of Each Product in Inventory", x = "Product", y = "Quantity Available") +
  theme_minimal()
ggsave("bar_chart_inventory_quantity.png", width = 7, height = 5)
```

Output



Question 2

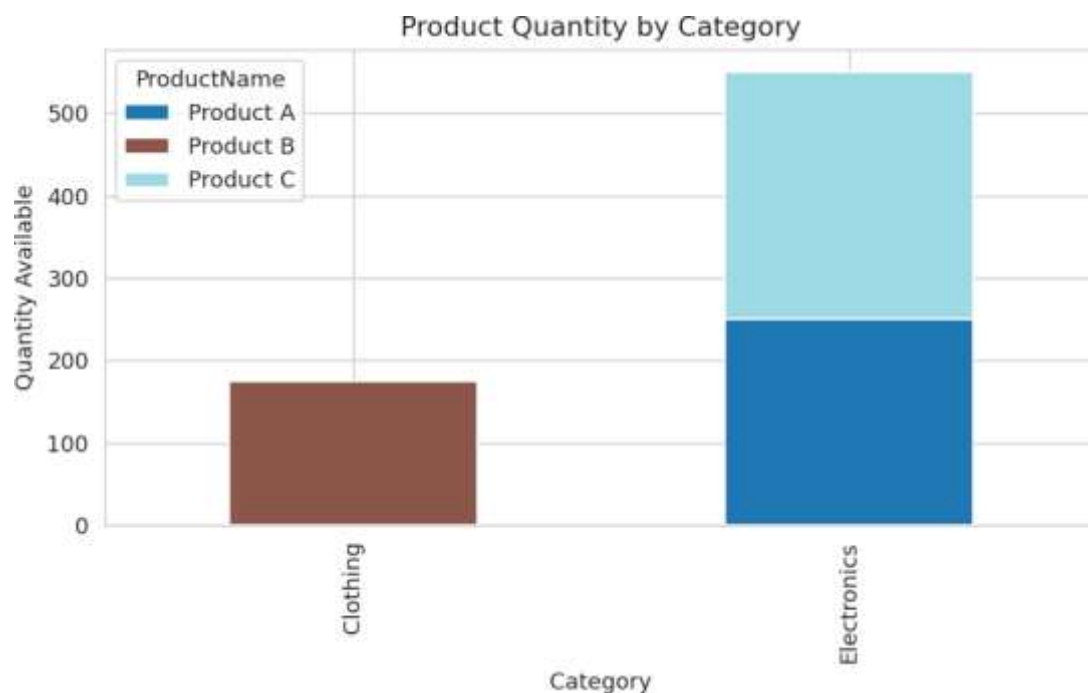
Generate a stacked bar chart to show the quantity of each product within different product categories.

R Code

```
# 2. Stacked bar chart - quantity within different categories
ggplot(df, aes(x = Category, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Product Quantity by Category", x = "Category", y = "Quantity Available") +
  +
```

```
theme_minimal()
ggsave("stacked_bar_quantity_by_category.png", width = 7, height = 5)
```

Output



Question 3

Build a table to show the inventory data for each product. Label the table elements.

R Code

```
# 3. Table - inventory data for each product
inv_table <- df %>% gt() %>% tab_header(title = "Product Inventory Data")
gtsave(inv_table, "table_product_inventory.png")
print(df)
```

Output

Product Inventory Data

ProductID	ProductName	Category	QuantityAvailable	Price
1	Product A	Electronics	250	20
2	Product B	Clothing	175	15
3	Product C	Electronics	300	18

Set 19: Survey Responses Analysis

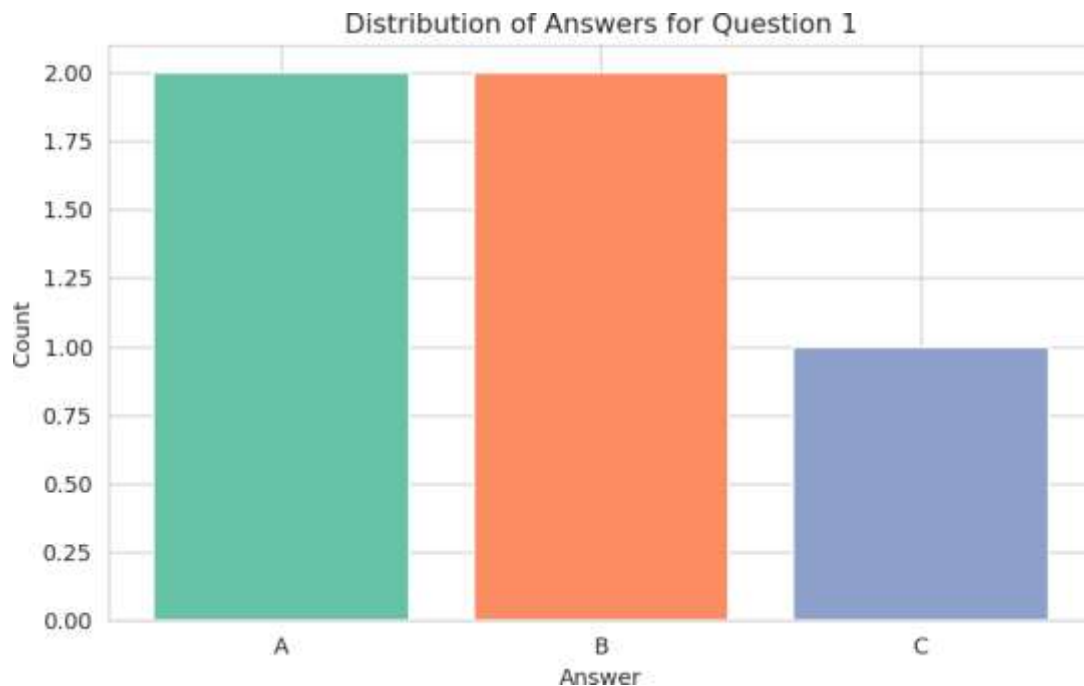
Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

R Code

```
# 1. Grouped bar chart - distribution of answers for Question 1
q1_counts <- df %>% count(Question1)
ggplot(q1_counts, aes(x = Question1, y = n, fill = Question1)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Distribution of Answers for Question 1", x = "Answer", y = "Count") +
  theme_minimal()
ggsave("grouped_bar_question1.png", width = 7, height = 5)
```

Output



Question 2

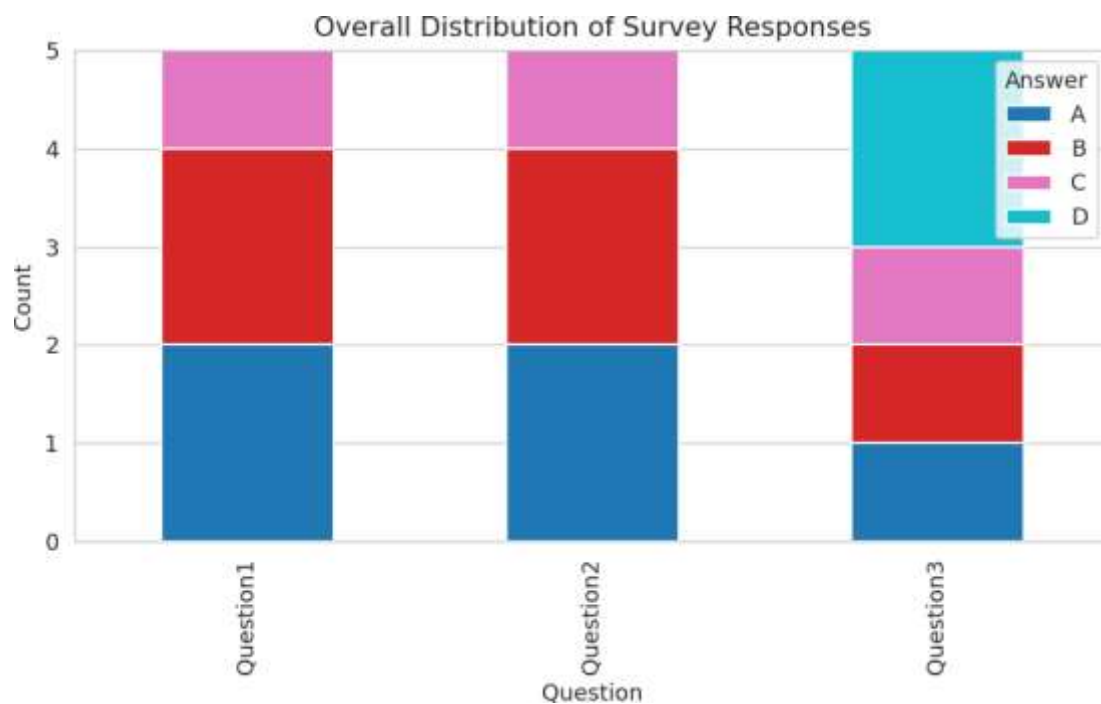
Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

R Code

```
# 2. Stacked bar chart - overall distribution of responses for all three questions
df_long <- df %>% pivot_longer(cols = c(Question1, Question2, Question3),
  names_to = "Question", values_to = "Answer")
ggplot(df_long, aes(x = Question, fill = Answer)) +
  geom_bar(position = "stack") +
  labs(title = "Overall Distribution of Survey Responses", x = "Question", y = "Count") +
  theme_minimal()
```

```
ggsave("stacked_bar_all_questions.png", width = 7, height = 5)
```

Output



Question 3

Build a table to show the survey response data for each survey. Label the table elements.

R Code

```
# 3. Table - survey response data for each survey
survey_table <- df %>% gt() %>% tab_header(title = "Survey Response Data")
gtsave(survey_table, "table_survey_responses.png")
print(df)
```

Output

Survey Response Data

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B
4	A	B	D
5	B	C	A

Set 20: Stock Analysis

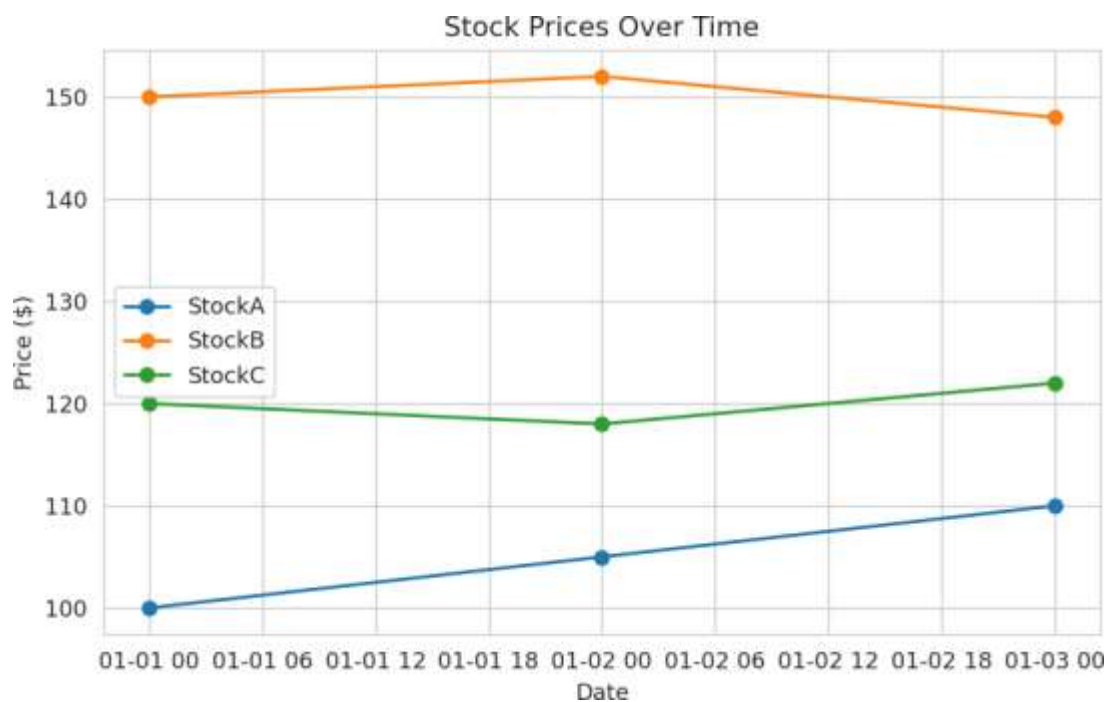
Question 1

Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

R Code

```
# 1. Line chart - stock prices of three companies over time
df_long <- df %>% pivot_longer(cols = c(StockA, StockB, StockC), names_to = "Stock",
  values_to = "Price")
ggplot(df_long, aes(x = Date, y = Price, color = Stock)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Stock Prices Over Time", x = "Date", y = "Price ($)") +
  theme_minimal()
ggsave("line_chart_stock_prices.png", width = 7, height = 5)
```

Output



Question 2

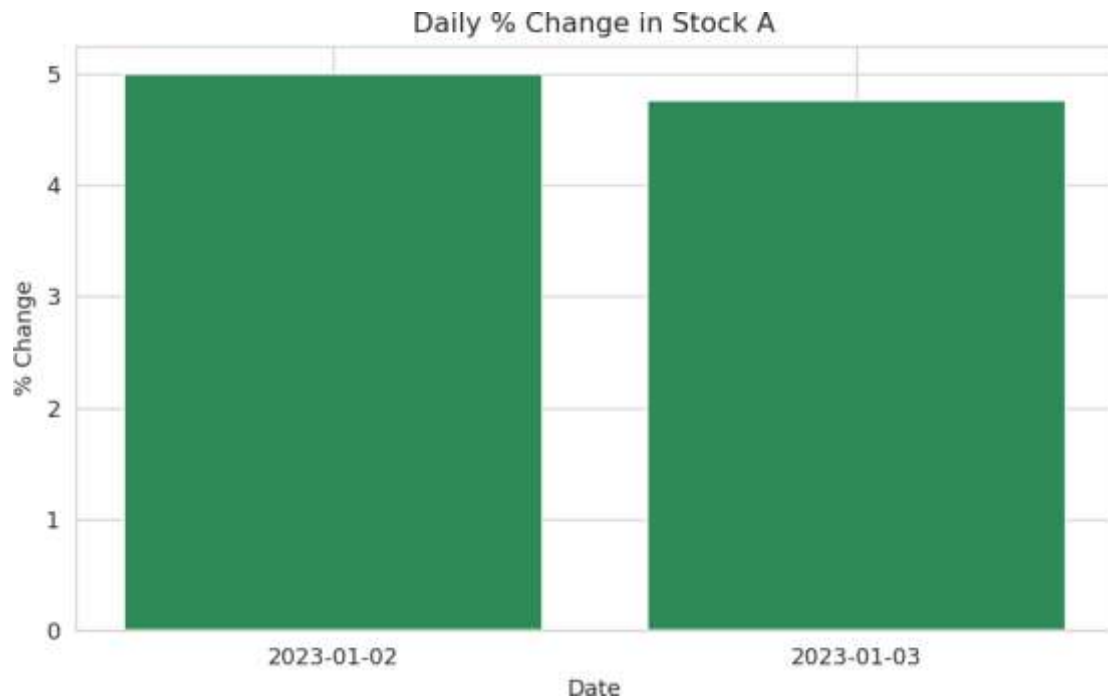
Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

R Code

```
# 2. Bar chart - daily percentage change in Stock A
df <- df %>% arrange(Date) %>% mutate(StockA_PctChange = (StockA / lag(StockA) - 1) *
  100)
```

```
ggplot(df %>% filter(!is.na(StockA_PctChange)), aes(x = as.character(Date), y =
StockA_PctChange)) +
geom_bar(stat = "identity", fill = "seagreen") +
labs(title = "Daily % Change in Stock A", x = "Date", y = "% Change") +
theme_minimal()
ggsave("bar_chart_stockA_pct_change.png", width = 7, height = 5)
```

Output



Question 3

Build a table to display the stock price data for each company over the given period. Label the table elements.

R Code

```
# 3. Table - stock price data for each company
stock_table <- df %>% select(Date, StockA, StockB, StockC) %>% gt() %>%
tab_header(title = "Stock Price Data")
gtsave(stock_table, "table_stock_prices.png")
print(df)
```

Output

Stock Price Data

Date	StockA	StockB	StockC
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122